SANJANA REDDY SINGIREDDY

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PROFESSIONAL SUMMARY

- Data Engineer with 4.5+ years of experience designing, developing, and optimizing enterprise-scale data pipelines, ETL/ELT frameworks, and cloud-based data platforms across healthcare, banking, and analytics domains.
- Strong expertise in Python, SQL, PySpark, Java, and distributed data processing with hands-on experience building scalable batch processing and data transformation workflows.
- Experienced in designing cloud data architectures using Azure (ADF, ADLS Gen2, Databricks, Synapse), AWS (S3, Glue, Redshift, RDS), and GCP (GCS, BigQuery) to support analytics and reporting workloads.
- Proven ability to develop and optimize ETL pipelines, data migrations, CDC frameworks, and data integration solutions using Airflow, Informatica, DataStage, and modern orchestration practices.
- Skilled in data warehousing and modeling including Snowflake, Teradata, dimensional modeling, star schema, data lakes, and lakehouse architectures for enterprise analytics platforms.
- Experienced in improving data platform performance through SQL optimization, query tuning, Spark transformations, Snowflake optimization, and pipeline performance enhancements, achieving measurable efficiency gains.
- Strong background in implementing data quality, validation, reconciliation, governance, metadata management, and automated testing frameworks to ensure reliable production data delivery.
- Adept at collaborating with cross-functional teams in Agile environments, leveraging CI/CD practices, Git, Azure DevOps, and cloud engineering best practices to deliver scalable and production-ready data solutions.

SKILLS

- **Programming & Query Languages:** Python, SQL, PySpark, Java, Shell Scripting, PL/SQL
- **Data Engineering & ETL/ELT:** ETL/ELT Development, Data Pipelines, Data Ingestion, Batch Processing, Data Transformation, Data Migration, Data Conversion, Change Data Capture (CDC), Workflow Automation, Apache Airflow, Informatica, SSIS, Talend
- **Big Data Technologies:** Apache Spark, Hadoop, Hive, HDFS, Sqoop, Databricks, Distributed Data Processing
- **Cloud Platforms:**
  Azure: Azure Data Factory, Azure Data Lake Storage (ADLS Gen2), Azure Databricks, Azure Synapse Analytics
  AWS: Amazon S3, AWS Glue, Amazon Redshift, Amazon RDS
  GCP: Google Cloud Storage (GCS), BigQuery
- **Databases & Data Warehousing:** Snowflake, Teradata, Oracle, SQL Server, PostgreSQL, MySQL, MongoDB, Data Warehousing, Dimensional Modeling, Star Schema, Snowflake Schema, Fact & Dimension Tables, Data Lakes, Lakehouse Architecture
- **Data Quality, Governance & Analytics:** Data Quality Frameworks, Data Validation, Data Profiling, Data Reconciliation, Data Lineage, Data Governance, Metadata Management, Data Auditing, PII Handling, Compliance Reporting
- **DevOps & Engineering Tools:** Git, GitHub, Jenkins, Azure DevOps, CI/CD Pipelines, Docker, Jira, Confluence, Maven
- **BI & Visualization:** Tableau, Power BI, Dashboard Development, Reporting Automation, Excel
- **Development & Collaboration:** Agile/Scrum, Kanban, Requirements Gathering, Cross-Functional Collaboration, Production Support

PROFESSIONAL EXPERIENCE

Illinois Department of Healthcare and Family Services – Remote, United States

Data Engineer | February 2024 - Present

- Tuned complex SQL workloads on billion-row Teradata tables by redesigning joins, execution plans, and query logic, cutting data preparation runtime from 2.5 hours to under 50 minutes.
- Directed enterprise data migration from Teradata EDW to AWS RDS Oracle using IBM DataStage, implementing scalable ETL workflows and source-to-target reconciliation for accurate production cutover.
- Built a reusable Python and SQL ETL framework with metadata-driven processing patterns, standardizing pipeline development and increasing engineering scalability.

- Designed enterprise-grade Apache Airflow workflows with dependency management, retry handling, logging, and automated reconciliation to support reliable production data operations.
- Established automated data validation frameworks using Great Expectations for schema checks, null validation, referential integrity, and business rule enforcement.
- Created a Python-based anomaly detection model using Isolation Forest to identify transactional irregularities and support proactive risk monitoring.

Key Achievements:

- Achieved 67% faster data preparation cycles, enabling quicker availability of enterprise datasets for reporting and analytics teams.
- Eliminated 60% of manual audit review effort by deploying automated anomaly detection capabilities for transactional data analysis.

Citibank – Boston, MA

Data Engineer | Mar 2023 - Jan 2024

- Created Azure Data Factory pipelines to ingest SQL Server and MongoDB data into ADLS Gen2, establishing scalable cloud data lake workflows for analytics workloads.
- Implemented SQL Server CDC and incremental ingestion frameworks, accelerating daily processing cycles and reducing ETL runtime by 50%.
- Constructed PySpark transformation pipelines on Azure Databricks using Bronze, Silver, and Gold layers to deliver curated Snowflake datasets.
- Enhanced Snowflake dimensional models through clustering, partitioning, and query tuning, increasing analytical query performance by 35%.
- Established automated Python and SQL-based reconciliation frameworks to validate source-to-target accuracy and strengthen production data reliability.
- Streamlined deployment processes using Azure DevOps, Git, and CI/CD workflows, ensuring consistent pipeline releases across Agile delivery cycles.

Key Achievements:

- Delivered 50% faster ETL processing through CDC adoption and incremental pipeline architecture, improving availability of business-critical data.
- Increased reporting efficiency by 35% through Snowflake performance tuning and enhanced warehouse optimization strategies.

Legato Health Technologies – Bengaluru, India

Analyst | July 2020 – Aug 2021

- Built Python ETL pipelines to process 50K+ healthcare records monthly from CSV sources, achieving 98% data accuracy through automated transformation and validation.
- Developed FastAPI ingestion APIs with Pydantic validation for provider data, improving data reliability by 25% and reducing downstream processing errors.
- Optimized PostgreSQL queries across 500K+ provider records and developed 5+ Power BI dashboards for operational reporting and stakeholder analytics.
- Conducted automated data quality audits using Python and SQL, identifying 200+ duplicate and inconsistent records and supporting remediation efforts.

Key Achievements:

- Reduced manual data entry errors by 30% by implementing automated validation rules and data quality checks in the provider onboarding workflow
- Improved Power BI dashboard refresh times from 5 minutes to 2 minutes by optimizing SQL queries and enabling incremental refresh

EDUCATION

- **Master of Science in Data Science |** University of Massachusetts | Dartmouth, MA | 08/2021 – 01/2023